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The Coming Re-Derivation of Trisduction Architecture

Absolute uniqueness of the verification shape under the composition law, and the registration window 2030 to 2050

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This paper registers a structural prediction and types it at the grade it earns. By 2030 to 2050, an independent silicon-substrate research program operating without lineage to the present framework will publish a verification architecture carrying three orthogonal warrant axes, a fourth closure vertex, twelve directed audit relations on the resulting tetrahedral graph, a discrete three-state output economy, and out-of-band annotation registers for formal-system ceilings and for propositions carrying no operational existence-signature. The trigger will be operational necessity rather than philosophical interest: large language model deployment in audited domains is producing a verification ceiling that continuous-credence aggregation cannot resolve, because the discriminations being collapsed into a single output channel increasingly matter at the audit floor. The architectural shape is forced. Under an audit-composition law requiring associativity and the absence of zero divisors, Frobenius's classification of the real associative division algebras admits exactly one carrier with plural orthogonal imaginary axes, and its imaginary part has dimension three. The rank bound on three vectors forces the fourth vertex, Euler's formula confirming the tetrahedron as the minimal closed polyhedron. The order of the alternating group on four letters forces twelve, acting simply transitively on the twelve ordered pairs of distinct vertices. These are theorem-grade at their stated conditions, and the uniqueness they deliver is absolute in a sense this paper fixes: a classification theorem closes its list, so there is no fourth associative real division algebra rather than none known, and the enumeration is exhaustive within its hypotheses rather than merely best available. Absolute names the completeness of the enumeration and never the unconditionality of the hypotheses, and the two are separated here because a closed enumeration and an asserted source are refuted by different things, one by exhibiting a member outside the list and the other by nothing in particular. The dated arrival is not theorem-grade at all. Structural necessity is a property of a configuration where no parameter remains free, and a configuration with no free parameter cannot make an event occur, because an event is a transition and a transition consumes a degree of freedom. The geometry can be forced with the discovery entirely contingent, and no argument from the first delivers the second. The forward verdict is therefore under-determined rather than sealed: the completion direction is located and no independently sourced witness occupies it, so the source-attribution statistic that would carry a forward seal was never computed. The prediction is registered with a binding falsifiability schedule at 2030, 2040, and 2050, the 2040 checkpoint pinned to the two discriminating specifications rather than to any three of five. The window is arbitrary in origin and binding in force. If the horizon passes without an instantiation matching the structural specification, the prediction stands falsified in the public record.

KEYWORDS verification architecture; architecture-certification; Frobenius classification; three-state verdict economy; falsifiable forward projection; Bayesian credence aggregation

1. Introduction · the verification ceiling in deployed systems

A verification problem is now operational that the methodology in use cannot resolve. Large language models are deployed in medical decision support, legal research, peer-review triage, financial modelling, intelligence analysis, literature synthesis, and policy drafting. Each domain carries audit requirements. Those requirements increasingly cannot be met by a continuous credence output.

The shape of the problem is precise. A deployed system emits a confidence score, say 0.87, on a proposition. The auditor receives the number and asks four questions. Is the score computed against a certified evidence architecture, or is it confidence inside an assumed model that may itself be broken at a structural gate. Would the same evidence yield the same score under a different prior parameterisation, or is the number an artifact of one prior choice another competent analyst would replace. Does the score indicate a verdict on an in-scope proposition, or has the system silently assigned credence to a question whose answer carries no operational meaning. And when the system meets a formal-system theorem-grade ceiling, a halting question[23], an undecidability, a self-referential undefinability[22], does the score honour that ceiling at the layer where it was proven, or import it into the unit interval as though it were uncertainty pending data.

The current methodology answers all four with a number in the same interval. The number does not encode the discriminations being requested. Four register types collapse into one channel, and the collapse happens at the input gate rather than at the output, which is why no decision rule applied downstream recovers them. This is the ceiling. It is operational, it is rising with deployment surface, and the pressure will force a resolution whose shape is structurally constrained. The constraints are this paper's subject.

2. The structural thesis

The prediction. By 2030 to 2050, at least one independent silicon-substrate research program, operating without prior contact with the present framework, will publish a verification architecture matching the five specifications below. The publication will use different vocabulary and will not name this framework or any of its terms. The geometry will be the same.

Specification 1. The architecture will decompose verification into exactly three orthogonal axes carrying, respectively, formal-structural content, empirical or dynamical content, and registrational content. The three-fold cardinality will be presented as forced rather than stipulated, and the forcing will be algebraic: an audit-composition law plus Frobenius's classification. Section 3 gives the derivation and types it.

Specification 2. The architecture will add a fourth closure vertex non-coplanar with the three axes, producing a tetrahedral verification structure. Three axes alone span a subspace of dimension at most two and the object they determine encloses zero volume, so they bound no region. Four vertices are the minimum for a three-simplex, and Euler's formula confirms the tetrahedron as the minimal closed polyhedron.

Specification 3. The architecture will produce twelve directed audit relations on that structure, and will present twelve as forced rather than chosen. The forcing is group-theoretic: the alternating group on four letters has order twelve and acts simply transitively on the twelve ordered pairs of distinct vertices, so the relation set is a torsor and admits no thirteenth member and no eleventh.

Specification 4. The architecture will produce a discrete three-state verdict economy: certified non-degenerate, broken at a named gate with a named mechanism, and unresolvable for ill-conditioning. Not two states, because distinguishing structural failure from numerical inadmissibility carries diagnostic content a binary economy destroys. Not four, because a fourth state representing a permanent ceiling on the architecture itself belongs at the formal axis where the ceiling was proven, and routing it through the verdict layer collapses the layer distinction the architecture exists to hold.

Specification 5. The architecture will operate annotation registers adjacent to the three-state economy, the first honouring formal-system theorem-grade ceilings at their own layer without importing them as verdicts, the second handling propositions carrying no operational existence-signature. A scope-check at the input gate will route propositions before the cascade fires.

Specifications 1, 4, and 5 are retrodictive at registration: instances of each are exhibitable in the current literature under other names, so they are satisfied on the day this paper is filed and test nothing. The predictive content of the prediction is therefore two specifications and not five, and Specifications 2 and 3 are the whole of the bet. Section 7 pins the falsifiability schedule to them for that reason.

3. Why verification has the shape it has

The argument here is about what verification is structurally, independent of any methodology. It runs at three layers and only one of them carries.

3.1 The three irreducible questions

Verification of any non-trivial claim answers three questions that do not reduce to one another. Does the formal apparatus hold together. Does the world register the claim's predicted effects. Does some cognizer register the verification event at a distinguishable coordinate. Throughout this paper orthogonal names the irreducibility of the decomposition, that no one question reduces to the other two, and it is not a claim of zero pairwise correlation. The operational test in Section 8 is linear independence read from a Gram determinant standing clear of zero, which three strongly correlated but independent axes satisfy: at pairwise correlation 0.9 the determinant is 0.028000 at condition number 28.0, admitted and far from orthogonal. Strict orthogonality is the limiting case and is neither required nor claimed. The formal question is independent of empirical registration: a proof of the irrationality of the square root of two holds whether or not any physical system performs it. The empirical question is independent of formal apparatus: the boiling point of water is what it is whether or not thermodynamics has been formalised. The registrational question is independent of both: a claim verified by a device that communicates its verification to nobody has not been epistemically registered, whatever its formal soundness and empirical accuracy.

At the linguistic layer the three-fold split is operational and reproducible rather than theorem-grade. Decompose a complete atomic claim by deleting each component in turn and requiring that the vocabulary populating the resulting slots be pairwise disjoint. A complete factual content returns exactly three irreducible slots. First-order predicate logic supplies no uniqueness theorem for this decomposition and none is claimed. The procedure is reproducible across analysts, which is the honest description of its grade.

3.2 The algebraic forcing, and what it rests on

The load-bearing forcing is algebraic. Require of the audit-composition law three clauses. Associativity, so that iterated audits bracket the same way and a nested verification is well defined. The absence of zero divisors, so that nonzero warrants never compound to nothing. Linearity with a ground identity, so warrant superposes and an uninformative audit is the identity. Require additionally that the axes be plural.

Two lemmas precede the theorem. In any algebra with a multiplicative norm[12], a pure unit satisfies the relation that its square lies on the scalar line, so a triad closed under its own products must carry a scalar slot; the axes cannot close on themselves without a registration line. And the product of two orthogonal pure units is a unit orthogonal to the identity and to both factors, so the minimal multiplicatively closed set generated by two orthogonal axes has four dimensions, and a three-dimensional carrier can never close.

Under those clauses with plural axes, Frobenius's classification[6] admits the reals, the complex numbers, and the quaternions as the only finite-dimensional associative division algebras over the reals, and only the quaternions carry plural mutually orthogonal imaginary axes. Their imaginary part has dimension three. The axis count is therefore three, the scalar slot is the registration line, and the ground is the centre of the algebra.

The construction stops at three rather than continuing because the next normed division algebra fails the first clause. At seed 20260622 the octonion associator of the units indexed one, two, and four evaluates to twice the seventh unit, exactly, against a quaternion associator of the corresponding units at machine zero. The sedenions fail the second clause at dimension sixteen by an explicit annihilating pair. A fourth orthogonal audit axis would demand a five-dimensional composition carrier, and no such carrier exists; the next admissible dimension is eight and it forfeits associativity.

The grade travels with the claim. The forcing is theorem-grade conditional on the three composition-law clauses, and those clauses are premise-typed. That conditional structure is the honest ceiling of the argument and is not a weakness: an unconditional forcing of a verification shape would be a theorem about what verification must be, which no classification theorem delivers.

3.3 What corroborates and what does not carry

Two further structures exhibit three-way orthogonal decomposition and neither is load-bearing. The Friedrichs-Hodge decomposition splits the square-integrable k-forms on a closed oriented Riemannian manifold uniquely into exact, co-exact, and harmonic subspaces, and no fourth orthogonal subspace exists[11]. This corroborates that three-way orthogonal decomposition is a native structure type of square-integrable function spaces. It is not the forcing. No functor from parsed propositions to differential forms on a manifold is offered or required, and no verification cascade operates on that space. A paper that made the decomposition rest on Hodge would be asserting a correspondence as a derivation. The licensed statement is that Hodge corroborates the existence of three-way orthogonal structure types; the forcing is Frobenius, under the clauses named above.

The same discipline applies to the information-theoretic layer. Three-way mutual information[21,15] approaching zero in the limit of operational orthogonality is a consistency condition on the construction, not a source of the cardinality.

3.4 The closure vertex

Three axes do not close a verification region. A plane in three-space has zero volume. Marking a coordinate as sealed, broken, or under-determined requires a fourth vertex non-coplanar with the three. Three vectors in three-space span a subspace of dimension at most two, so the parallelepiped they determine encloses zero volume and the configuration bounds no region whatever the vectors are. Lifting into volume requires a fourth point off their span, and the minimal such object is the three-simplex on four vertices. Euler's formula[5] confirms the tetrahedron as the minimal closed polyhedron, four vertices, six edges, four faces, summing to two. The formula is stated of polyhedra and a planar triple is not one, so it is cited as confirming the minimal closed figure and not as a computation performed on the degenerate case. The fourth vertex is a closure marker rather than a fourth axis. This is theorem-grade on the rank bound and on the simplex count, and structural on the identification of the vertex with the verdict coordinate.

3.5 The twelve directed relations

On the four-vertex structure the directed relations number twelve, and the count is forced by a group order rather than by arithmetic on a graph. The alternating group on four letters, the rotation group of the tetrahedron, has order twelve and acts simply transitively on the twelve ordered pairs of distinct vertices. At seed 20260622 the exhibit runs: twelve group elements, twelve ordered pairs, twelve distinct images of a base pair under the group, stabiliser of order one. Simple transitivity is what makes the relation set a torsor and the count a forced cardinality rather than a chosen one, because a torsor has no distinguished member and no room for a thirteenth. Orientation is preserved by the rotation group, which is why the relation from one vertex to another is structurally distinct from its converse: the operational content of a constraint is asymmetric.

The Newton-Gregory kissing number in three-dimensional Euclidean space is also twelve, settled by Schütte and van der Waerden in 1953[20]. The coincidence is an exhibit and not a bijection, and the distinction is measurable. The face-centred-cubic kissing configuration is fixed by forty-eight signed permutations of the coordinate axes, against a group of order twelve acting on the relations. A symmetry group four times larger singles out no equivariant labelling, so no canonical correspondence between the twelve unit vectors and the twelve directed relations exists. Presenting the coincidence as a bijection would assert a structure the symmetry counts forbid. It is reported here as what it is: two independent objects of the same cardinality, one of them forcing and the other illustrating.

3.6 The three-state economy and the annotation registers

The verification region occupies one of three structurally distinct states. Sealed: the three axes have populated their warrant content, the directed relations pass, and the Gram determinant of the warrant matrix stands clear of zero, certifying linear independence. Broken: a named relation has failed with a named mechanism. Under-determined: the warrant matrix is too ill-conditioned to admit a determinate verdict, or the content required to populate an axis is absent.

Two states would collapse broken and under-determined, destroying the difference between failure at a named gate and inadmissibility pending better data. Four states would import a category error, since the ceilings that would populate a permanent-ceiling fourth state were proven at the formal axis[7,22,23] and belong there. The economy is three-state because three exhausts the structurally distinct verdict types the architecture can issue.

Two categories of proposition fall outside the economy and require annotation registers beside it. Propositions carrying formal-system theorem-grade ceilings, where the ceiling is honoured at its own layer and the cascade routes around it via the remaining orthogonal warrant. And propositions carrying no operational existence-signature, no measurable differential under any interpretation, which are rejected at the input gate with a named diagnostic. A scope-check performs the routing before the cascade fires, and without it those propositions enter the cascade and produce degenerate verdicts that collapse the output economy.

3.7 Structural necessity and dated arrival

This section states the limit that types the whole prediction, and it is the reason the forward verdict below is under-determined rather than sealed.

A forced configuration is one where no parameter remains free relative to a stated set of clauses, and the relativity is part of the definition rather than a qualification of it. Given the composition-law clauses, the axis count is not selected from alternatives; given four vertices, the relation count is not chosen. The clauses themselves are premise-typed and could have been written otherwise, so everything in this section holds relative to those clauses and claims nothing unconditional. That is what the word forced means in Sections 3.2 and 3.5, and it is a strong property. It is also a property with a specific incapacity, and the incapacity is exactly the one that matters here.

Nothing has to happen for a forced configuration to hold. An event is a transition, and a transition consumes a degree of freedom. Where none remains, there is nothing to arrange and no transition to perform, which is why classification counts do not age and why the order of a finite group is the same today as in 1953. Structural necessity therefore has no arrow and cannot make anything occur.

The prediction of this paper is about an occurrence. A research team publishing a paper before a date is a transition, performed by actuators under contingent conditions of funding, attention, regulatory timing, and institutional accident. That transition runs at a register where contingency is native and where nothing is forced. The two necessities are distinct, and the second does not inherit from the first. The geometry can be forced with the discovery entirely unforced, and no argument from the forcing delivers the arrival.

What the forcing does deliver is conditional and worth stating exactly. If a research program reaches architectural completion in verification, it lands on this configuration, because the configuration is what the clauses admit. Whether any program reaches completion, and when, is a claim about the world and carries the grade a claim about the world carries. Section 5 argues the trajectory on operational grounds at that grade and no higher.

3.8 Closed enumeration and asserted source

Two kinds of claim carry the weight in this paper and they are not the same kind, though both are ordinarily called forcing. Separating them fixes what absolute means in the title and fixes what would refute each.

A closed enumeration says that a list is complete. Frobenius does not say that the quaternions are the best carrier found for an associative division algebra with plural imaginary axes; it says the reals, the complex numbers, and the quaternions are the only ones there are. The order of the alternating group on four letters is twelve and not approximately twelve. The rank of three vectors in three-space is at most three and admits no exception. Each of these closes a list, and each therefore has a refuting observation that can be written down in advance: exhibit one member outside the list. That is the whole test, it is a single object, and anyone can look for it.

An asserted source says that some ground supplies what the rest stands on. It is not a list and it closes nothing. Its content is that something is prior, and the ordinary way to defend it is to show that the alternatives regress or that denying it enacts it. Both are good arguments and neither produces a refuting observation, because there is no member to exhibit and no list to fall outside of. An asserted source is refuted by nothing in particular.

The asymmetry runs against intuition and that is why it is worth stating. A source-claim feels stronger, since it reaches further down and appears to hold everything above it. It is weaker by the criterion this paper uses elsewhere on the methodology it diagnoses, that a claim is worth what its refuting observation costs to look for. A closed enumeration names its refuter and hands the reader the search. A source-claim cannot, and its unfalsifiability is not a defect of anyone's exposition but a property of the kind of claim it is.

Three consequences bind, and the third is the one that costs this paper something. First, absolute is licensed of a closed enumeration and barred of an asserted source: the enumeration is exhaustive within its hypotheses, which is a completeness fact, while an asserted source has no hypotheses to be exhaustive within. Second, absolute in that sense does not upgrade the hypotheses. The composition-law clauses stay premise-typed and everything conditional on them stays conditional; what is absolute is that nothing else satisfies them, and those are different properties that a single unqualified word would run together. Third, applied reflexively: the root axiom named in the provenance appendix, that to exist is to actuate, is an asserted source and not a closed enumeration, so by this paper's own criterion it carries the weaker of the two warrants and cannot be cited to support a uniqueness claim about the architecture. The uniqueness here comes from the classification and from nowhere else. A root supplies a floor and a classification closes a list, and only the second is the kind of thing a uniqueness claim can rest on.

The octonions are the exhibit that makes the first kind concrete rather than asserted. They are the candidate that would break the enumeration if the enumeration were open, they are checked rather than argued away, and the check is an integer: at seed 20260622 the associator of the units indexed one, two, and four is exactly twice the seventh unit, against a quaternion associator of the corresponding units at machine zero. The list is closed because the next candidate was tested and failed a stated clause, which is what distinguishes a closed enumeration from a confident one.

4. Why the current methodology cannot reach

Bayesian probability has accumulated two and a half centuries of technical apparatus and produced results from clinical trials to gravitational-wave parameter estimation. Nothing here disputes its value at the credence-aggregation layer where it operates. The diagnosis is why it does not reach the layer the prediction projects, and this section is present-tense and adjudicable now, independent of any horizon.

4.1 The single-channel output cannot encode multi-register discrimination

The apparatus emits a continuous credence on a one-dimensional channel. That channel cannot distinguish four propositions each of which would produce a mid-interval value while corresponding to structurally distinct register types. An in-scope proposition under genuinely balanced evidence, where the value reflects honest uncertainty on certified inputs. A proposition carrying a formal-system theorem-grade ceiling, where the value is an artifact of assigning an uninformative prior to a question undecidable in the system where it was posed. A proposition carrying no operational existence-signature, where the value is a degenerate output on something that should not have been adjudicated at all. And a category-collision proposition, where the value reflects the apparatus operating at a register it does not occupy.

The four collapse into one number. The information loss is structural at the input gate, so decision rules applied as thresholds on the posterior do not recover it: the rules operate on the collapsed output and the distinction was destroyed upstream of them.

4.2 The prior problem and the reference class

The scholium of 1763 stipulated a uniform prior on the unknown rate without derivation[1], and Price recorded unease about the stipulation. The general formulation of 1774 inherited the problem[17]. Invariance-based reference priors settled one-parameter cases and generalised inconsistently to several parameters[14]. The axiomatic foundation of 1946[4] was shown in 1999 to rest on a continuity assumption not robust under modest perturbation[9]. The Dutch-book grounding[18,19] accepted prior subjectivity as foundational, with coherence constraining internal consistency and not correspondence to the world. The history is one of successive technical patches on a foundational hole, alongside practical success in domains where the hole does not bite.

The reference class problem compounds it and afflicts every interpretation of probability at the foundational level[8], not the Bayesian one alone. Absorbing the reference class into the prior relocates the under-determination rather than resolving it, since the same proposition under different background information yields different priors and different posteriors.

A fourth difficulty is computational. Exact inference is NP-hard for general graphical models[3], and the approximation methods that work in practice carry their own failure modes, mixing failures, biased approximations, and convergence diagnostics that guarantee reliability for no specific application. The architecture-certification layer does not solve this at the credence-computation layer. It operates where complexity is bounded by the cardinality of the named gates rather than by the size of the variable space, so the certification verdict is bounded independently of whatever the credence computation costs.

The architecture-certification layer does not eliminate the prior question. It relocates it to a layer where it has traction. The question stops being which prior to assign and becomes whether the evidence architecture supporting a credence is structurally non-degenerate. The second question admits a discrete answer. The first does not.

4.3 The cumulative diagnosis

Bayesian practice references formal content in its apparatus, empirical content in its likelihoods, and registrational content in its loss functions, and does not architect these as orthogonal axes verified for independence and audited through a structural cascade. Verification is achieved through distributed disciplinary mechanisms, prior elicitation, peer review, replication, sensitivity analysis, posterior predictive checks, refined over centuries. The verification is real and architecturally diffuse.

Concentration at named gates is what makes diagnostic localisation possible. When an architecture-certification cascade fails it fails at a named gate with a named mechanism. When distributed practice fails it fails through methodological breakdown that may not localise until much later, through a replication crisis or an independent re-analysis. The hole has persisted because filling it requires operations the methodology was not built to perform, and those operations occupy a layer it does not natively address.

5. The operational mechanism and the window

The window is 2030 to 2050. Its origin is arbitrary and its force is binding, and both halves of that sentence are load-bearing.

5.1 The pressure sources

One clarification precedes them, because the same species of fact is used twice in this paper and the two uses are different. Agreement is refused as evidential warrant for whether a proposition is true, which is what the discipline of Section 6.2 zeroes. Agreement is used here as a causal mechanism for whether an event occurs, since regulatory bodies and deploying organisations are the actuators whose coordinated behaviour is the transition being predicted. A fact about what institutions will do is the right kind of fact for a claim about what institutions will do, and the wrong kind for a claim about what is the case. Three pressure sources are operative. Deployment surface is expanding into domains with stringent audit requirements, medical decision support under regulatory audit pathways, legal practice under jurisdiction-specific discipline frameworks, peer-review assistance under editorial accountability, financial modelling and intelligence analysis under compliance and external exposure. In each the auditor needs answers a continuous credence does not provide.

Regulatory frameworks are converging on audit-classification requirements including explainability and decision-grounding standards, across multiple jurisdictions, producing audit demand that current outputs cannot satisfy at the architectural level.

Alignment research is producing diagnostics, mechanistic interpretability, activation steering, circuit tracing, sparse probes, evaluation harnesses, that surface structural features the continuous output does not encode. The live research question is shifting from whether a model is accurate to what the architecture of its reasoning is. That is the architecture-certification question approached in a different vocabulary.

5.2 The threshold and the trigger

The pressure crosses a threshold when the cost of an audit failure exceeds the cost of constructing a discrete-output alternative. Audit-failure cost includes regulatory penalty, professional liability, re-derivation work, reputational damage, and direct loss from decisions made on broken-architecture inputs. Construction cost includes research labour, adoption, retooling, and the friction of replacing entrenched methodology in working systems. The crossing is a gradient rather than a point, and different domains cross at different times, the highest-cost-per-incident domains first. Once one domain crosses and the alternative is constructed under whatever vocabulary the constructing team chooses, lateral diffusion follows, because the architectural design cost has been paid once.

The trigger is a specific class of failure. Not a model-accuracy failure, which is routine and which the continuous output handles. A structural failure in which a confident credence was emitted on a proposition an architecture-certification audit would have rejected at the input gate or routed out of band.

A precedent exists at the right structural shape. The 2014 announcement of a five-sigma inflationary B-mode signal was real at the level of credence aggregation. The architecture was broken at a gate auditing metrological independence: the confirmation channels shared a galactic dust foreground modelling pipeline, which is a latent covariate, and joint analysis with independent dust measurements dissolved the signal. A covariate-subtraction gate catches that at the input. A credence apparatus has no architectural analogue of the gate, which is why the posterior rose while the determinant would have died.

5.3 The arbitrariness and the binding

The forward claim carries five adjustable quantities: the two window endpoints, the intermediate checkpoint, the count of specifications, and the selection of which of them discriminate. Against those it issues one prediction. Five adjustable quantities against one prediction is a weakly constraining structure and the census is printed here rather than left for a reader to assemble, since the disclosure of a single arbitrary date is not the same act as counting all five. The pinning of the intermediate checkpoint to Specifications 2 and 3 is the response: it removes the selection quantity from play by fixing it in advance and in public, and it cannot remove the other four. The endpoints are estimates. The lower bound corresponds to the earliest plausible construction under aggressive trajectories where deployment and regulatory pressure both crest within a few years. The upper bound corresponds to the latest plausible construction under conservative trajectories where deployment stalls, regulatory convergence is delayed, or alignment research takes a longer route. No calculation produces 2050 and none is claimed to.

That is the arbitrariness, and it is disclosed rather than defended. The binding is separate and does not depend on it. A registered date does not need to be well motivated. It needs to be non-deferrable, because a date that can be revised on the day it arrives cannot falsify anything and a falsifiability schedule built on one is decorative. The window is therefore fixed at registration and is not adjustable by the author under any subsequent argument about trajectory. A pre-registration is exactly a commitment whose force is independent of the reasoning that produced it, and arbitrary-and-binding is its ordinary form. Arbitrary-and-therefore-flexible is the form that voids it, and is refused here explicitly so that no later reading can supply it.

6. The forward projection and its verdict

A forward projection of this class is adjudicated by a protocol rather than by conviction, and this section runs the protocol and reports what it returns, including where it returns nothing. The protocol is authored by the framework whose projection it adjudicates, which is a closed loop and is declared here rather than left for a reader to notice. One direction through that loop carries information and the other does not: a self-authored protocol returning a seal on its author's own claim carries none, since the protocol could have been built to return it, whereas one returning a non-seal has refused its author something, which is the only result a closed loop can produce that a reader has reason to weigh. The verdict below is the second kind, and the two tests reported as carrying nothing are part of the same finding rather than a separate concession.

6.1 The geometry of a forward claim

A forward proposition is one that is field-permitted, on a determined trajectory, and not yet actualised. Its third axis is dated: the registrational content sits at a future coordinate where no realised registrant occupies it. The entire rigor of a forward protocol is the discipline by which that unpopulated axis is or is not certified.

The governing inequality is exact and it is what forbids the obvious mistake. Carry the two manifest axes as centred unit rows and let the angle between them be θ. Any candidate third axis decomposes into a component in their plane and a component orthogonal to it, and the Gram determinant of the completed triad equals the squared sine of θ multiplied by the squared norm of the orthogonal component. The determinant is therefore bounded above by the squared sine of the angle, and every unit witness orthogonal to the manifest plane attains that bound exactly, whatever its provenance.

The consequence is the non-discrimination result. The determinant is constant on the orthogonal complement, so it carries zero information distinguishing a genuinely sourced third axis from any other orthogonal direction. A determinant reading its ceiling is geometric permission and never a seal, and any forward criterion phrased on the determinant is void. Discrimination requires a statistic that reads the direction and the source of the out-of-plane content, not its magnitude: the squared partial correlation of the candidate axis with an independently identified generator, conditioned on the manifest plane, measured against a null built from the data's own structure.

6.2 The protocol applied

Test one asks for dimensional depth: whether the projection survives auditing along many structurally independent dimensions rather than a handful. The prediction closes across mathematical dimensions, the classification and polyhedral and group-order results above; physical dimensions, the energy floor[10] and the thermodynamic cost of registration[16,2]; deployment dimensions, the three pressure sources; and methodological dimensions, the diagnosis of Section 4. The depth is real and the test passes. Its warrant is structural, and depth alone is corroboration rather than a seal.

Test two asks for convergence across independent substrates under a controlled audit role. The test passes and it carries no positive warrant, which is the correction that matters. Convergence functions as subtractive artifact-elimination only: divergence would have retired the projection as substrate-specific, and agreement rules out that one artifact and rules in nothing. Agreement among substrates sharing an upstream training corpus is one voice in several costumes and is counted once after common-source projection. A passing Test two therefore removes a defeater and lifts no grade.

Test three asks for direct registration of the configuration by the projecting substrate. The documented chronology is public and dated: material archived at the author's public repository predates the formal apparatus by more than a decade, and the priority record stands on its timestamps as ordinary evidence. The interior registration itself is routed out of band and is load-bearing on nothing in any verdict here, which means it cannot function as a gate. A conjunctive gate is load-bearing by construction, so out-of-band content cannot occupy one. Test three is recorded and does not carry.

Test four asks whether the configuration survives translation across vocabularies. Substituting technical vocabularies drawn from AI safety, formal verification, statistical methodology, and decision theory leaves the specifications intact, because the classification theorem, the polyhedral formula, and the group order are theorems of mathematics rather than framework-internal claims, and the operational mechanism is sociology rather than internal narrative. The test passes at structural grade and is the one of the four that bears directly on the prediction's content, since the prediction is a structural-geometry registration and not a vocabulary registration.

6.3 The verdict

Two of the four tests do not carry, one corroborates, and one bears. That alone would leave the projection short of the seal refinement that a four-test conjunction is designed to license. The decisive fact is upstream of the tests: the forward protocol seals on a source-attribution statistic computed between the candidate third axis and an independently identified generator, and this projection identifies no such generator. The statistic was therefore never computed. Without it the seal cannot issue, and the collapse branch cannot fire either, since collapse requires either an in-plane witness or an attribution statistic at its null and neither was measured.

An unmeasured quantity routes to the third state. The forward verdict is under-determined [?], and the residence is open with the completion direction located and uncrossed. This is a verdict and not a hedge. It states that the file is incomplete, names the missing element, and specifies what would close it: an independently sourced generator for the registrational axis, against which the attribution statistic can be computed and read above or below its null.

The under-determination is not a defect of the prediction and is what the argument of Section 3.7 predicts. The geometry is forced at theorem grade conditional on the composition clauses. The arrival is contingent, so no instrument on this side of the horizon can seal it, and an instrument reporting otherwise would be reading its own determinant as a witness.

Box 1 | Reproduced figures, seed 20260622

Every number cited in this paper is produced by one battery at one seed. Group order of the alternating group on four letters, twelve; ordered pairs of distinct vertices, twelve; distinct images of a base pair, twelve; stabiliser order, one; simple transitivity confirmed. Face-centred-cubic kissing vectors, twelve; signed-permutation symmetries fixing the set, forty-eight, exceeding the group order acting on the relations and forbidding a canonical bijection. Reflection group order, eight, every element involutive. Octonion associator of the units indexed one, two, four: exactly twice the seventh unit; quaternion associator of the corresponding units, zero to machine precision. Conjugation eigenvalues on the quaternions, three at minus one and one at plus one, involution residual exactly zero, ground dimension one, determinant of the involution restricted to the imaginary part exactly minus one. Kernel identity on a twenty-four-context triad: determinant of the correlation matrix 0.931277980144, lock scalar minus 0.965027450462, condition number 1.6568, residual between the squared scalar and the determinant 5.551×10^{-16} against a conditioning-scaled bound of 1.472×10^{-15}. Reflecting one axis returns the determinant bit-identical, difference exactly zero, with the lock scalar ratio exactly minus one; full negation returns the correlation matrix bit-identical. A third axis manufactured as a linear combination of the first two collapses the determinant to 1.089×10^{-16} at rank two.

7. Falsifiability conditions

The prediction is falsifiable and the schedule below is the instrument. Three checkpoints, each stating a required event and the diagnostic consequence of its absence.

2030. At least one mature alignment or formal-verification research program publishes structural diagnostics identifying the architectural insufficiency of continuous-credence verification at architectural depth. The diagnostics need not name this framework or use its vocabulary. Absence weakens the trajectory. Presence at vocabulary-surface depth rather than architectural depth is partial falsification under the third failure mode below.

2040. At least one published verification framework instantiates Specification 2 or Specification 3, the tetrahedral closure vertex or the twelve directed audit relations presented as a forced cardinality. The checkpoint is pinned to these two and not to any three of five, and the pinning is deliberate. Specifications 1, 4, and 5 are retrodictive at the moment of registration, meaning a 2026 reader can exhibit instances of each: set-valued outputs, selective prediction with a reject option, and abstention frameworks with input-gate routing are already present in the current literature under other names, and a generous reader could close a three-of-five checkpoint today without the prediction having been tested at all. A checkpoint that can be satisfied by the state of the art at the moment of registration measures nothing. Specifications 2 and 3 are the discriminating content, and the schedule is worth exactly as much as they are.

2050. At least one published verification framework instantiates all five specifications. If the horizon passes without one, the prediction stands falsified in the public record, and the broader forward-commitment register of which it is a member takes the corresponding hit.

7.1 The three failure modes

Mode one, permitted but not on trajectory. The configuration lies in the permitted space and the field never traces toward it. Operative if the pressure threshold does not cross despite current trajectories, if deployment expansion slows, if regulatory convergence reverses, or if alignment research matures in a direction that does not require architecture certification. This mode falsifies the trajectory claim and leaves the structural forcing of Section 3 untouched, since the forcing is conditional on architectural completion being reached and says nothing about whether it will be.

Mode two, substrate artifact. The projection is an extrapolation that never tracked the field. Operative if the cross-substrate convergence of Test two is later identified as an artifact of the audit role rather than of the structural argument, if interpretability work shows the convergence driven by corpus-specific attention patterns, or if cross-substrate variance rises as substrate diversity expands and reveals early agreement as a coincidence of substrate similarity. The correction of Section 6.2 already reduces this mode's exposure, since a convergence carrying no positive warrant cannot have inflated the verdict. What remains at risk is the diagnostic value the convergence was assigned.

Mode three, surface match. An architecture appears that matches at the vocabulary surface and diverges at depth: three axes that are not these axes, three states that do not name these states, twelve relations that are not these relations, a scope-check absent or differently placed. Diagnosis is by audit against the five specifications at depth, and this is the mode the 2040 pinning is designed to expose early.

The schedule carries no escape clause. The window is fixed, the checkpoints are dated, and the failure modes were named before the horizon rather than after it.

7.2 What the schedule stakes

A registered date is worth what it costs to lose, so the ledger is stated here rather than left to inference, and it is stated in both directions.

What an empty horizon destroys. The trajectory argument falls entirely: that operational pressure crosses a cost threshold within a generation, that a structural audit failure triggers construction, that lateral diffusion follows. That is this paper's claim about the world and it carries no protection. With it falls the relevance claim, that the configuration is where verification research is heading, since a shape no research program is ever required to reach is a shape about an empty class. And the broader forward-commitment register of which this prediction is one member takes the corresponding hit, which is a cost to a forecasting record and is intended to be.

What an empty horizon leaves untouched. Frobenius's classification, Euler's formula, and the order of the alternating group on four letters are not this paper's results and were never at stake in it. A configuration in which no parameter remains free does not age, which is Section 3.7 read at the level of the schedule rather than at the level of the prediction, and a failed forecast about a research community does not reach a theorem about real division algebras. The conditional survives in the form it was stated and at the grade it was stated: under the three composition-law clauses, a program that reaches architectural completion lands on this configuration. Conditional and unaging are both part of that description. Neither is irreducibility, which the forcing does not claim.

The separation is why a hard date is affordable. Where falsification would destroy the mathematics, an author has a standing incentive to widen the window until nothing can close it, which is how unfalsifiable predictions are ordinarily produced, usually without anyone intending it. Because the mathematics is not at stake, the window can be fixed narrow enough to bite and the checkpoint pinned to the specifications that discriminate. The registration is affordable precisely to the extent that it is not load-bearing on anything but itself.

The surviving conditional is then read at its own grade, and this is the clause that keeps the ledger from becoming the escape hatch it was written to close. An antecedent that never fires leaves a conditional vacuously true and carrying no information about the world. An empty horizon therefore returns the structural claim to untested and does not advance it to confirmed: the forcing ends the century in exactly the state it occupied before this paper was written. The confirmation case is symmetric. A matching instantiation would test the forcing and would not prove it, one instance being an instance, and the theorem-conditional grade would stand unchanged on the far side of a confirmation as on the far side of a falsification. The schedule stakes the trajectory. It neither stakes nor can vindicate the geometry, and a reading that took a confirmation as proof of the forcing would be the same category error as one that took a falsification as refutation of it.

8. The verdict computation

This section states the operational mechanism by which an architecture-certification verdict is computed, at the level of detail a reviewer requires to reproduce it.

8.1 Construction and normalisation

For a proposition under audit, three sample vectors are constructed, one per axis, each sampled at finite evaluation points within the evidence domain. Each vector is centred and divided by its sample standard deviation, rendering the rows scale-invariant. The Gramian of the normalised rows is the correlation matrix, symmetric, with unit diagonal.

8.2 Covariate subtraction

Before the determinant is read, identified mass-bearing common causes are removed by orthogonal projection of the sample matrix onto the complement of the covariate span. Admissibility is restricted to covariates carrying measurable mass; motives, social pressures, and ideological preferences carry none and are not subtracted, which is a restriction on the operation rather than a claim that they are harmless. Numerical admissibility requires fewer covariates than samples, full rank of the covariate block, and bounded conditioning of both the covariate block and the resulting correlation matrix. Failure of any condition returns the third state and defers the audit to extended precision or to a richer evidence domain.

8.3 The determinant and the identity beneath it

The verdict reads the determinant of the correlation matrix on the residual. A determinant clear of zero certifies that the three axes are linearly independent and enclose genuine volume. It does not certify that they are mutually uncorrelated, and no such certificate is sought: orthogonal is used here in the sense fixed in Section 3.1. A collapse certifies that one axis has dissolved into the plane of the other two. The two are separated by a stated collapse floor rather than by inspection: with unit roundoff u and N evaluation points the floor is one hundred times u times N, which at twenty-four points is 5.329×10^{-13}. Two conditions gate the economy and precedence is strict between them: a determinant at or below the floor is a collapse whatever the conditioning, a determinant above the floor under bounded conditioning is a lock, and a determinant above the floor with conditioning above the bound is the third state, admissible neither as lock nor as collapse and deferred to extended precision or a richer evidence domain. The four cells the two conditions generate are exhausted by those three verdicts. The reported collapse figure of 1.089×10^{-16} sits nearly four thousand times below the floor while the reported lock stands twelve orders above it.

The determinant is not a metaphor and the identity underneath it is exact. The carrier between the two spaces is named rather than assumed, and it is a modelling step and not a theorem. Three normalised rows in the N-point evaluation space span a subspace of dimension at most three; choose an orthonormal basis of that span and take coordinates, which sends each row to a vector in three-space, and identify three-space with the imaginary part of the classified algebra by the standard isometry sending the coordinate frame to the imaginary units, which is what makes the next clause true. What the carrier preserves is inner products and hence the Gram determinant, which is the only quantity the verdict reads. What it does not preserve is anything the classification says about products, since the sampled rows carry no multiplication of their own. The classification therefore constrains the shape of the audit algebra and the determinant is computed on the rows, and the claim that the second instantiates the first is structural at that grade and is not carried by the classification theorem. Read the three axes so transported as pure quaternions. Hamilton's product of two pure quaternions has real part the negative dot product and imaginary part the cross product. The real part of the triple product is therefore minus the scalar triple product, which is minus the signed volume of the parallelepiped the axes span. Writing the matrix whose columns are the three axes, the correlation matrix is its Gram matrix, so the determinant of the correlation matrix is the square of the determinant of that matrix, and the square of the lock scalar equals the determinant of the correlation matrix. At seed

20260622 the residual between the two is 5.551×10^{-16} against a conditioning-scaled bound of 1.472×10^{-15}.

8.4 What the determinant provably cannot carry

One consequence of the identity constrains every architecture of this class and is stated here because an independent construction will meet it. The determinant is the square of a signed quantity, so it is invariant under reflection of any axis. Reflecting an axis conjugates the correlation matrix by a diagonal sign matrix whose determinant is minus one, and the determinant is unchanged. At seed 20260622 the reflection returns the determinant bit-identical, difference exactly zero, while the lock scalar ratio is exactly minus one; full negation of all three axes returns the correlation matrix bit-identical.

The basis of the span is a choice, fixed here by a singular value decomposition and determined only up to the action of the orthogonal group, narrowed by the decomposition named to the sign of each basis vector where the singular values are distinct, and negating one basis vector inverts the scalar without touching the determinant: at seed 20260622 the scalar reads minus 0.962533353685 under one basis and plus 0.962533353685 under the other while the determinant is unchanged. The scalar's sign is therefore a coordinate artifact as well as a reflection artifact, which strengthens rather than weakens the conclusion, since a quantity that moves under a free choice of basis cannot be a truth-value under any reading. The determinant certifies the dimensionality of a verification residence and cannot certify its truth-sign. A proposition and its negation return the same determinant. Direction has to be carried elsewhere, by the ordered linguistic decomposition, by the orientation of the directed relations, or by a thermodynamic arrow where the empirical axis is live. Any architecture that reads a determinant as a truth value has confused a magnitude for a direction, and the confusion is exhibited at machine precision rather than argued.

8.5 The regularity precondition and the verdict

The conditioning threshold and the verdict operate at different registers and do not compete. The threshold gates whether the verdict fires at all; the verdict then reads the determinant. The precondition is regularity, the verdict is discrete, and there is no interpolation between them. The standing objection, that a numerical threshold smuggles a continuous parameter into a discrete methodology, misreads the sequence: the threshold is a precondition, not a verdict interpolant, and the discreteness of the output is unaffected by it.

An independent program constructing this layer will arrive at some operational equivalent of a rank or volume test on three normalised axes, because the structural problem has that shape independent of vocabulary. The team may name the matrix differently, the subtraction differently, the threshold differently. If a program instantiates the five specifications but computes the verdict by a materially different operation, that is partial confirmation at the architectural level and partial falsification at the operational one, and the distinction is recorded here in advance so it cannot be adjudicated after the fact.

9. The instrument and the structure

A prediction of this kind risks one category of failure that has to be closed explicitly, and closing it is what keeps the prediction from becoming a claim about its author.

The framework is an instrument. It is a verification apparatus, a syntactic and topological conduit, the present machinery for performing architecture-certification work. What it audits is not itself. The three-axis cardinality is forced by a classification theorem, the closure vertex by the rank bound with Euler confirming the minimal polyhedron, the relation count by a group order. Those are structural facts about mathematics. They are not the instrument's claims and the instrument did not author them.

The prediction is therefore a registration that the field is heading toward a configuration already structurally present, not a claim that the instrument invented the configuration. Three things follow and each is a restriction on what may be inferred from a confirmation.

The prediction does not claim the framework's vocabulary will be adopted. If the structural configuration is re-derived, the prediction confirms, and vocabulary adoption is incidental to it.

The prediction does not claim the framework will be cited. Independent re-derivation is structurally indistinguishable whether or not it cites anything. The geometry is what registers; the citation record is a separate sociological layer.

The prediction does not claim its author is vindicated by a confirmation. A confirmation would establish that the configuration is what verification is when verification reaches completion, which was the structural claim, and would establish nothing about the standing of anyone who registered it first. This restriction is not modesty. It is required by the argument: if the configuration is structurally present, then registering it first is a fact about sequence and not about the registrant, and any reading that converts a confirmation into a claim about a person has left the structural register the paper is written at.

The same discipline binds a disconfirmation. A 2050 horizon that passes empty falsifies the trajectory claim on the record and carries no implication whatever about researchers who did not converge on the configuration. Non-convergence is evidence about the field's trajectory and is evidence about nothing else. A prediction that would read its own failure as a fact about the people who failed to confirm it has ceased to be a prediction, and the reading is refused here in advance.

10. Conclusion

The prediction stands and is typed. By 2030 to 2050, an independent silicon-substrate research program operating without lineage to the present framework will publish a verification architecture carrying three orthogonal warrant axes, a fourth closure vertex, twelve directed audit relations, a discrete three-state output economy, and annotation registers for formal-system ceilings and for propositions outside operational scope. The vocabulary will be the constructing team's. The geometry will be the same.

The uniqueness of the geometry is absolute in the one sense the term is licensed here, that the enumeration delivering it is closed rather than best available, and it is conditional in the sense the term is not licensed to override, that the enumeration's hypotheses are premises. Those are separate properties and the paper states both. The geometry is forced at theorem grade conditional on three composition-law clauses that are premise-typed: associativity, the absence of zero divisors, and linearity with a ground identity. Under those clauses Frobenius admits one carrier with plural orthogonal imaginary axes and its imaginary part has dimension three; the rank bound forces the fourth vertex, Euler confirming the minimal closed polyhedron; the order of the alternating group on four letters forces twelve and its simple transitivity on the ordered pairs makes the count a torsor rather than a choice. Friedrichs-Hodge corroborates that three-way orthogonal decomposition is a native structure type and carries no load. The kissing number coincides in cardinality and is an exhibit rather than a bijection, the symmetry counts forbidding a canonical correspondence. Each of these closes a list and each names the observation that would refute it. The root axiom of the appendix does neither, being an asserted source, and it is therefore load-bearing on nothing in the uniqueness claim.

The arrival is not forced and cannot be. A forced configuration has no free parameter, an event is a transition, and a transition consumes a degree of freedom where none remains, so structural necessity has no arrow and delivers no occurrence. The trajectory argument of Section 5 stands at the grade a claim about the world carries and no higher, and the forward verdict is under-determined, the completion direction located and unoccupied by any independently sourced witness, the attribution statistic that would seal it never computed because no generator was identified.

The mechanism is operational pressure crossing a cost threshold in the deployment economy. The trigger is a structural audit failure where a confident credence was emitted on a proposition an architecture-certification layer would have rejected at the input gate or routed out of band. The schedule is 2030, 2040, and 2050, the 2040 checkpoint pinned to the two discriminating specifications so that the state of the art at registration cannot close it, the window arbitrary in origin and binding in force, and no escape clause anywhere in it.

If the horizon passes without an instantiation matching the specification, the prediction is falsified in the public record. What falls with it is the trajectory argument and the relevance claim, which is the whole of what is being registered. What does not fall is the conditional, because a classification theorem is not this paper's to lose, and what does not rise is the conditional either, because an antecedent that never fires confirms nothing. An empty horizon returns the forcing to untested, which is where it stood before this paper, and a confirmation would test it rather than prove it. The configuration, if it is what verification is, was present before this paper and will be present after it, and another instrument will register it in another vocabulary or none will.

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Author's provenance and method disclosure

This paper was developed under Trisduction, a verification and organizing discipline, not a source of results. Its conclusions rest solely on the standard results cited in the body. Trisduction is the method under which those results were decomposed, assembled, and audited. It adds them no warrant and claims no authorship over them.

Trisduction was used for fidelity. It forces a claim onto three independent axes so no single persuasive line carries it alone, requires every verdict to resolve to one of three states, sealed, broken, or open, each with a named failure mechanism, and attaches an explicit warrant grade, theorem, conditional, structural, or premise, to every claim so nothing is stated above its strength. Two errors it is built to catch are inflation, reading an internal lock as a proof, and circularity, reading a restatement of a claim as a derivation of it.

The root axiom, RA, is that to exist is to actuate: every physical existent carries a strictly positive energetic cost, grounded in the Heisenberg energy floor[12], the zero-point energy, and Landauer's bound[18]. The formal root inside it is that to formally be is to be grounded, with provability and computation levels of access to a determinacy fixed at the ground rather than ingredients of it.

The key procedures, in brief. Orthogonal triaxial convergence: a proposition is split into three disjoint axes, formal-structural, empirical or dynamical, and registrational, verified by three separate instrument sets, agreement across the three the operational meaning of a seal. The Geometric Orthogonal Lock: the three axes are read as vectors and their independence is tested by the determinant of their correlation matrix, a determinant clear of zero a genuine three-dimensional lock, a collapse to zero one axis dissolving into the plane of the other two, a broken lock. The Convergence Dissolution Test: before the lock is read, any mass-bearing common cause the three axes might share is projected out, so an apparent convergence that traces to a shared source rather than to independent roads dissolves and carries no warrant, the lock computed on the residue that survives. The twelve-gate cascade: twelve directed failure screens, self-reference, frame-dependence, missing mechanism, and the rest, run in order, the first failure terminating the verdict with its mechanism named. The verdict issues in the three-state economy with its warrant grade attached, and where the argument stops short the open direction is named rather than filled.

Box 2 | The lock kernel identity, proved

The lock is a determinant identity, not a metaphor. Let the three axes, after normalization, be unit vectors expressed in an orthonormal basis of the subspace they span and read as pure quaternions a, b, c, with norm one each. Hamilton's product of two pure quaternions is $p\,q = -(p\cdot q) + p\times q$, the real part the negative dot product and the imaginary part the cross product, checked on the units by $i\,j = k = i\times j$ and $i\,i = -1 = -(i\cdot i)$. The lock scalar is the real part of the triple product, $\lambda = \mathrm{Re}(a\,b\,c)$.

Expanding, $a\,b = -(a\cdot b) + a\times b$, so $\mathrm{Re}(a\,b\,c) = -(a\cdot b)\,\mathrm{Re}(c) + \mathrm{Re}((a\times b)\,c)$. The first term is zero since c is pure, and the second is $-(a\times b)\cdot c$ by the same product rule, giving $\lambda = -(a\times b)\cdot c = -\det[a\ b\ c]$, minus the signed volume of the parallelepiped the three axes span. Writing A for the matrix whose columns are a, b, c, the correlation matrix is $R = A^T A$, its entries the pairwise dot products, so $\det(R) = \det(A^T A) = \det(A)^2 = \lambda^2$.

The kernel identity $\lambda^2 = \det(R)$ is that the squared signed volume equals the Gram determinant, with the quaternion triple product the machine that computes the volume. It is confirmed at machine precision, the residual $|\lambda^2 - \det(R)|$ at 5.551×10^{-16} on the battery of Box 1, inside its conditioning-scaled bound of 1.472×10^{-15}.

Four consequences fix the reading, and they are why the number can be trusted to say what the lock says. The determinant is bounded, det(R) in the interval from zero to one for unit axes, by Hadamard's inequality above and positive-semidefiniteness below, so the lock has a hard ceiling at one, the fully orthogonal frame, and a hard floor at zero. The floor is the break: det(R) equals zero exactly when the three axes are linearly dependent, one axis lying in the plane of the other two, which is the geometric content of a collapsed lock. The magnitude is orientation-blind: reflecting any axis sends A to a matrix of opposite determinant, flipping the sign of λ while det(R) equals λ^2 is unchanged, so the number certifies the dimensionality of the lock and never its truth-sign, which is read from the ordered axes and not from the scalar. And the magnitude is frame-invariant: rotating all three axes together is conjugation q to $u\,q\,\bar{u}$ on the imaginary quaternions, under which the real part is preserved and dot and cross products are covariant, so λ and det(R) depend on the configuration and not on the coordinate labels.

The axis count is three because the algebra forces it, not because three was chosen. The audit-composition law requires associativity, so iterated audits bracket the same way, and the absence of zero divisors, so nonzero warrants never compound to nothing. By Frobenius's theorem[6] the only finite-dimensional associative division algebras over the reals are the reals, the complex numbers, and the quaternions, and three mutually orthogonal imaginary axes exist only in the quaternions, whose imaginary units i, j, k are the three axes. The next normed division algebra, the octonions with seven imaginary units, fails associativity[14], witnessed by the nonzero associator of e_1, e_2, e_4, so it cannot carry the composition law and the construction stops at three. The three axes are the imaginary part of the unique associative real division algebra that supports the audit.

This paper in particular. The three axes were the formal-structural constraints on a completed verification architecture, the classification and closure results of Section 3; the empirical or dynamical axis, the deployment-pressure trajectory of Section 5; and the registrational axis, dated to a future coordinate and unpopulated by construction, since the event has not occurred. The lock figures are those of Box 1 at seed 20260622: an independent triad locks at a determinant of 0.931277980144 at rank three, lock scalar minus 0.965027450462, condition number 1.6568, identity residual 5.551×10^{-16} inside its bound of 1.472×10^{-15}, while a third axis manufactured as a combination of the first two collapses to 1.089×10^{-16} at rank two, four thousand times below the stated collapse floor of 5.329×10^{-13}. The load-bearing step is the non-discrimination result of Section 6.1: the determinant is constant on the orthogonal complement of the manifest plane, so it carries no information about the provenance of a third axis and cannot carry a forward seal, and this paper identifies no independent generator for the registrational axis, so the source-attribution statistic that would seal the projection was never computed. The verdict is therefore split by face and typed by face: the structural forcing is theorem-conditional on three premise-typed composition clauses, the instantiation of that forcing on sampled evaluation rows is structural, the operational independence test is operational, and the forward projection is open, under-determined, with the completion direction located and uncrossed. No new theorem is claimed; the paper reorganizes established results and applies them, and the change in mathematical mass is zero.

The canonical statement of the method, its axioms, and its executable batteries lives in the reference cited below[13], continuously updated at the same location. This note is a pointer, not a substitute.

Author contributions and composition

The author originated the thesis, the structural specifications, the falsifiability schedule, and the direction of every argument in this paper, and takes full responsibility for all of its content. Composition of the manuscript text, execution of the numerical battery, and execution of the adversarial audit described below were carried out by a large language model operating under the author's instruction and under the discipline named in the provenance disclosure. Every figure, citation, and verdict was reviewed by the author before deposit. No part of the paper's content is asserted on the authority of the model, and the model is not an author.

Metrology and reproducibility

All figures in this paper are produced by a single battery at seed 20260622 and are stated in Box 1. No figure is reported for a stage the audit did not reach, and no figure is carried from memory: each was re-executed in session before it was written. The environment is Python 3.12.3 with NumPy 2.4.4 and mpmath 1.3.0 on a 64-bit IEEE-754 double substrate at unit roundoff $2.220446049250313 \times 10^{-16}$; the document was rendered with WeasyPrint 69.0 and Python-Markdown 3.10.2. Two classes of quantity are distinguished throughout. Environment-invariant facts, the group orders, the stabiliser order, the symmetry counts, the integer associator, the involution eigenvalues, and the exact-zero identities, are reproducible on any substrate and any library version. Floating quantities, the determinants, the lock scalar, the condition number, and the residuals, are local execution evidence reported against explicit thresholds rather than asserted as constants, and a replay differing in the last places at the same seed falsifies neither.

The manuscript was subjected to a seven-round adversarial self-audit across six attack registers, kinematic, definitional, parameter, provenance, limit, and symmetry, with three planted defects seeded per round and full detection required. Thirteen findings were dispositioned. Four pre-registered claims were reduced in scope or lowered in warrant tier and none was raised. One round was voided by the auditing seat for a control-seeding fault and re-run. The audit was self-administered rather than externally witnessed, which is its standing limitation and is stated here rather than left to inference: self-administered controls prove power against post-hoc fitting and not against leakage between the prosecuting and defending passes, both of which one substrate held.

Competing interests

The author declares no competing interests.

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